Activity 4 Guide

Course: Semiconductor Process Fundamentals Masterclass

 Course code: C716
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 Activity: 4 - Integrate Film Deposition and CMP
 Alignment: LO1
 Duration: 55 minutes

Goal

Trace deposition, fill and CMP interactions and design a verification plan that protects both planarity and electrical performance.

Scenario

A dual-damascene copper level meets line-resistance targets at wafer centre but fails near the edge with dishing, erosion and suspected barrier continuity loss.

What You Need

- Cross-section cards and film/CMP worksheet
- Legacy diagrams for barrier/seed, copper fill, slurry, polishing and post-CMP clean
- Evidence template

Detailed Step-by-Step Procedure

1. Label the dielectric, barrier/liner, seed, copper fill and overburden layers on the supplied cross-section.
2. Identify which layer prevents diffusion and which layer enables electroplating continuity.
3. Compare centre and edge thickness/planarity observations and classify dishing, erosion or incomplete clear.
4. Trace how PVD coverage or electroplating voids can change the apparent CMP result.
5. Choose two CMP inputs to review: downforce, platen speed, pad condition, slurry chemistry, flow or polish time.
6. For each input, state the expected effect on removal rate, selectivity and within-wafer uniformity.
7. Add a post-CMP clean and contamination-control check to the integration plan.
8. Define metrology at three points: pre-CMP overburden, post-CMP topography and electrical line/via continuity.
9. Write a hold/release rule based on planarity plus electrical evidence.

Acceptance Criteria

[ ] The layer stack and each module's function are correctly labelled.

[ ] The analysis distinguishes deposition/fill causes from polishing causes.

[ ] The release rule uses both physical and electrical evidence.

Reflection Questions

1. Why can a visually planar surface still fail electrical verification?
2. Which measurement best distinguishes dishing from incomplete copper clear?

Evidence to Submit

- Completed worksheet.

- Completed evidence template with file names or screenshots.
- The activity output named with the A04- prefix.
- A final release/hold/recommendation statement supported by measurements.

Troubleshooting

- If a workbook opens in protected view, save a local copy before editing blue input cells.
- If formulas do not update, enable automatic calculation and press Ctrl+Shift+F9 (Windows) or Command+= (macOS Excel).
- If a diagram becomes crowded, separate the process flow from the evidence/control loop; cross-reference them with numbered hand-offs.
- If the conclusion is uncertain, state what is known, what is not known, and the next measurement that can resolve it.